Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry

A PROJECT REPORT

Submitted in partial fulfillment of the requirements

for the award of the degree of

B.TECH

IN

ELECTRONICS & TELECOMMUNICATION ENGINEERING

by

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2026–2027

CERTIFICATE

This is to certify that the project report entitled "**Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry**" has been completed by **Soham Govind Kedari (2025201008)**, **Aryan Nandkumar Kupekar (2025201009)**, and **Amar Ankush Ware (2025201016)** under the guidance of Prof. Milind Paraye for the award of the degree of **B.Tech in Electronics & Telecommunication Engineering** from the University of Mumbai.

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PROJECT REPORT APPROVAL CERTIFICATE

This is to certify that the project report entitled “Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry” by Soham Govind Kedari, Aryan Nandkumar Kupekar, and Amar Ankush Ware is approved for the award of the degree of B.Tech in Electronics & Telecommunication Engineering from the University of Mumbai.

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Abstract

The project titled “Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry” is designed to monitor the weight of goods carried by a vehicle and respond when the measured load exceeds a configured threshold. The system uses four half-bridge load cells installed on a goods-carrying platform. Their wiring is combined into a single sensor bridge connected to an HX711 load-cell amplifier and analogue-to-digital converter. An STM32 Blue Pill microcontroller reads the weight measurement and compares it with the configured overload limit of 60 kg.

When the measured weight is 60 kg or less, the relay remains OFF, allowing the vehicle to start. When the measured weight exceeds 60 kg, the controller switches a 5 V relay through a BC547 transistor driver. The relay interrupts the line associated with the vehicle ECU and fuel pump, preventing the vehicle from starting. An ESP-01 Wi-Fi module is included to support data transfer to a Python-based website. The current website functionality includes live weight readings and an overload alert.

The hardware has been built and the following functions have been tested: live weight readings on the website, overload detection above 60 kg, relay switching to prevent the vehicle from starting, and the overload alert on the website. The report describes the confirmed hardware arrangement and leaves unverified implementation details and future additions clearly identified rather than assuming results.

Keywords: STM32 Blue Pill, load cell, HX711, ESP-01, IoT, overload detection, relay driver.

Acknowledgement

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Chapter 1

Introduction

Goods-carrying vehicles are used to transport materials and products across a wide range of applications. Excessive loading can create safety risks and may place additional stress on vehicle components. A practical way to support safer operation is to measure the load and provide a response when the measured weight exceeds a defined limit.

This project, "Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry," combines load measurement, microcontroller-based decision-making, relay control, and web-based monitoring. Four half-bridge load cells are mounted on the goods-carrying platform. Their combined wiring is connected to an HX711 module, which conditions and digitizes the small signal produced by the load-cell bridge. An STM32 Blue Pill processes the readings and checks them against a threshold of 60 kg.

When the weight is 60 kg or less, the relay remains OFF and the vehicle can start. If the weight exceeds 60 kg, the STM32 switches the relay through a BC547 transistor driver. The relay interrupts the line associated with the vehicle ECU and fuel pump, preventing the vehicle from starting. An ESP-01 Wi-Fi module supports sending live data to a website developed using Python.

Scope of the Project

The project is intended as a weight-based safety system for goods-carrying applications and is not limited in this report to a single vehicle category. The implemented threshold described here is 60 kg. The report documents the actual hardware arrangement and the tests confirmed by the project team. It does not claim that the system has been validated on every vehicle type.

Problem Statement

The project addresses the need to monitor the weight of goods and provide a defined response when the measured load exceeds a configured threshold. It also aims to make weight information available through a website for remote observation.

Chapter 2

Literature Review

This chapter outlines the technical concepts relevant to the project. A formal source-by-source literature survey has not been supplied by the project team at this stage; therefore, this chapter is limited to background concepts directly related to the implemented system. Verified references can be added to the bibliography when selected by the team.

2.1 Weight Measurement and Load Sensing

A load cell converts a mechanical load into an electrical signal. Strain-gauge load cells change their electrical resistance when the sensing element deforms under load. A bridge arrangement converts small resistance changes into a differential voltage. Because the resulting signal is small, an instrumentation amplifier and high-resolution conversion stage are commonly used before a microcontroller processes the reading.

In this project, four half-bridge load cells are installed on the goods-carrying platform and combined through wiring into one bridge arrangement connected to a single HX711 module. The HX711 provides signal amplification and analogue-to-digital conversion for the controller. The report does not assign individual measurements to the four cells because they are combined into one measurement channel.

2.2 IoT Monitoring and Vehicle Safety

Internet of Things (IoT) systems connect physical sensing and control equipment to networked applications. A microcontroller can acquire sensor readings and share selected data through a Wi-Fi module with a remote website. This makes it possible to view measurements without being physically beside the sensing hardware.

The ESP-01 Wi-Fi module is included in this project to support communication with a Python-based website. The website currently provides live weight readings and an overload-alert feature. The exact network protocol and server-side implementation details have not yet been finalized by the team and are therefore not specified in this report.

Chapter 3

Project Objectives

- To measure the total weight of goods placed on a goods-carrying platform using four half-bridge load cells.
- To combine the load-cell wiring into a single bridge measurement connected to an HX711 module.
- To use an STM32 Blue Pill microcontroller to read the measurement and compare it with a 60 kg overload threshold.
- To operate a 5 V relay through a BC547 transistor driver when the measured weight exceeds the configured threshold.
- To prevent the vehicle from starting during an overload condition by interrupting the line associated with the vehicle ECU and fuel pump.
- To send live weight data to a Python-based website using an ESP-01 Wi-Fi module.
- To provide live weight monitoring and an overload-alert feature on the website.
- To test the implemented hardware functions and document only the results confirmed by the project team.

Chapter 4

Theory

4.1 Half-Bridge Load Cells

A half-bridge load cell contains strain-sensitive elements that change resistance when mechanical force is applied. In a bridge arrangement, resistance changes produce a small differential voltage related to the applied load. The four load cells in this project are installed on the platform and combined by wiring into a single measurement bridge.

4.2 HX711 Load-Cell Amplifier and ADC

The HX711 is a load-cell interface integrated circuit that amplifies a low-level differential signal and converts it into digital data. The STM32 reads the converted value and uses the project's calibration and weight-processing logic to obtain the displayed weight. The exact calibration procedure and numerical calibration constants have not been provided, so they are not stated here.

4.3 STM32 Blue Pill

The STM32 Blue Pill board is the main controller in the system. It receives measurement data from the HX711, evaluates the weight against the 60 kg threshold, and controls the relay driver. It also participates in the data path used to provide readings to the ESP-01 module.

4.4 BC547 Relay Driver

The STM32 GPIO operates at a logic level that is not intended to supply a relay coil directly. In this design, a BC547 transistor is used as a switching driver, with resistors and a protection diode. The transistor controls the current path for the 5 V relay coil. The diode is placed to suppress the inductive voltage transient generated when the relay coil is switched off.

4.5 Relay-Based Starting Inhibit

The relay provides an electrically controlled contact that changes state in response to its coil. In this project, the relay is arranged to interrupt the line associated with the vehicle ECU and fuel pump during overload. At or below 60 kg, the relay remains OFF and the vehicle can start; above 60 kg, the relay switches and prevents starting.

4.6 ESP-01 Wi-Fi Module

The ESP-01 is a Wi-Fi module used to support wireless data transfer. In this project it is intended to provide live weight readings and overload information to the Python-based website. The specific communication protocol has not yet been confirmed and is not assumed in this report.

Chapter 5

System Design

The system is organized into sensing, processing, actuation, power-supply, and wireless-monitoring sections. The load cells measure the weight on the platform. The HX711 conditions and digitizes the combined bridge signal. The STM32 Blue Pill processes the measurement and applies the configured threshold. The BC547 transistor driver switches the relay, while the ESP-01 supports data communication with the website.

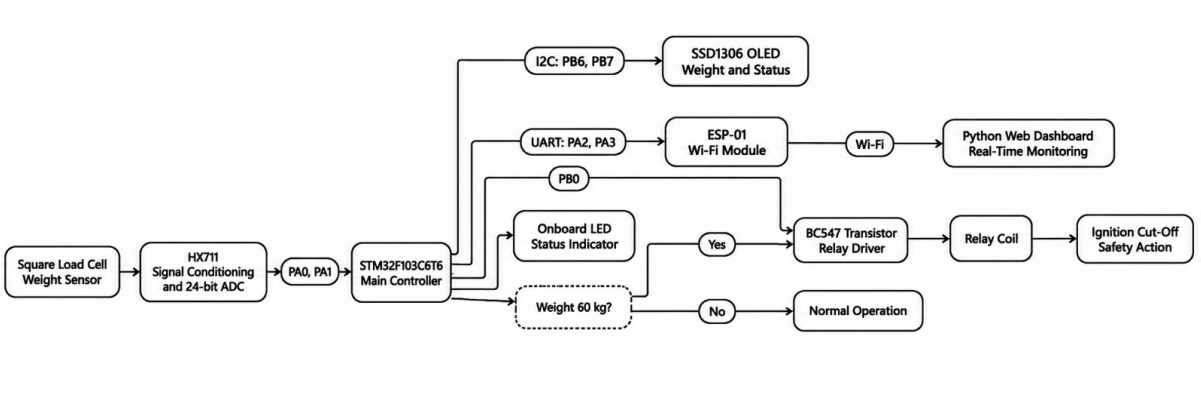


Figure 5.1: Block diagram of the proposed system

5.1 Main Hardware Components

Table 5.1: Main hardware components

Component	**Role in the system**	**Confirmed detail**
Four half-bridge load cells	Sense total load on the platform	Combined through wiring
HX711	Amplifies and digitizes the bridge signal	One module
STM32 Blue Pill	Reads weight and controls response	Programmed using STM32CubeIDE
BC547, resistors, diode	Relay driver stage	One BC547 transistor
5 V relay	Switches the starting-inhibit line	Controlled by STM32 via driver
ESP-01	Wi-Fi data transfer	Website uses Python
Separate 3.3 V regulator	Provides ESP-01 supply	Exact model not specified

5.2 Power Supply

The system operates using a 9 V DC input supplied by an adapter associated with the vehicle. A 5 V supply is obtained from the input using a buck converter. A second buck converter is used to step down the 5 V supply to 3.3 V, which is required for the ESP-01 Wi-Fi module. The regulated supply voltages are provided to the respective circuit components according to their operating voltage requirements.

5.3 Relay Operation

When the measured weight is 60 kg or less, the relay remains OFF, allowing the vehicle to start. If the weight exceeds 60 kg, the controller activates the relay, interrupting the ECU and fuel pump connection to prevent starting. This helps prevent vehicle operation under overloaded conditions and improves transportation safety.

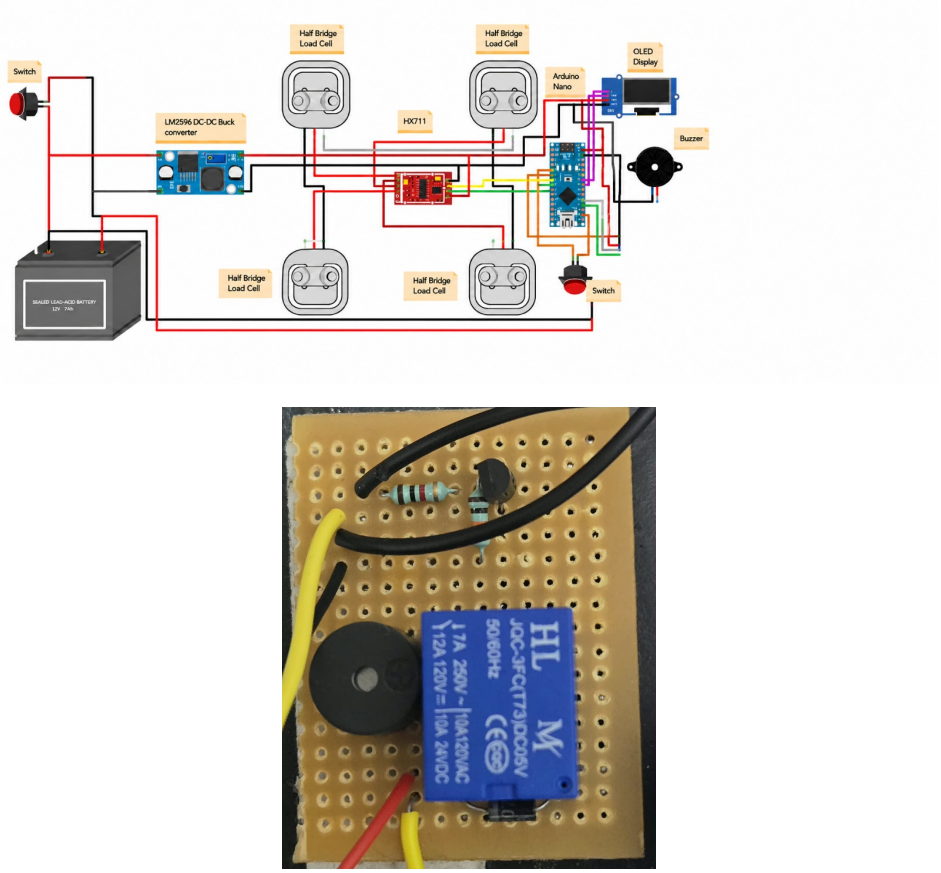


Figure 5.2: Hardware photograph and relay-driver circuit

Chapter 6

Softwares

The following software tools have been confirmed by the project team.

Table 6.1: Software tools used

Software / tool	**Use in the project**
STM32CubeIDE	Used to develop the STM32 microcontroller firmware.
STM32 Programmer	Used to program or flash the STM32 board.
Python	Used for the website that displays live weight data and includes an overload-alert feature.

6.1 STM32CubeIDE

STM32CubeIDE is an integrated development environment (IDE) used for developing, compiling, debugging and managing firmware for STM32 microcontrollers. It provides a common platform for writing embedded C programs and configuring the microcontroller peripherals required for a project. It helps developers create and maintain firmware that controls the operation of external hardware components.

In this project, STM32CubeIDE is used to develop the firmware responsible for processing the weight measurements obtained from the HX711 load cell amplifier. The STM32 microcontroller receives the measurement data and evaluates it against the predefined weight limit of 60 kg. Based on the measured weight, the firmware determines whether the goods-carrying vehicle is operating within the permitted load limit or whether an overload condition has occurred.

The firmware also manages the control signal provided to the relay driver circuit. When the measured weight exceeds the specified threshold, the microcontroller generates the required control signal to activate the relay arrangement. This allows the system to interrupt the connection between the vehicle ECU and the fuel pump, preventing the vehicle from starting under an overload condition.

STM32CubeIDE provides the development environment required to implement the control

logic and integrate the microcontroller with the connected hardware. It also supports the compilation and debugging process, helping in the development and verification of the embedded firmware.

6.2 STM32 Programmer

STM32 Programmer is a software utility used to program STM32 microcontrollers by transferring compiled firmware into their internal memory. It provides an interface for connecting to an STM32 device, identifying the connected microcontroller and performing programming-related operations.

After the firmware is developed and compiled using STM32CubeIDE, the generated program file must be transferred to the STM32 board. STM32 Programmer is used for this purpose. It allows the developed firmware to be loaded into the microcontroller so that the programmed instructions can be executed by the hardware.

In the proposed system, the firmware contains the instructions required to process the weight measurements, compare the measured weight with the predefined 60 kg limit and control the relay driver according to the detected condition. Programming the STM32 board makes these instructions available to the microcontroller, allowing it to perform its intended functions as part of the complete hardware system.

The programmer is therefore an important part of the development process because it provides the means to transfer updated firmware to the controller whenever changes or improvements are made to the program. It helps simplify firmware loading during development and testing.

6.3 Python Website

Python is a high-level programming language used in the development of the website for the proposed system. It provides programming features that can be used to develop web applications, process incoming information and present data in a user-friendly format.

In this project, Python is used to develop the website that provides a monitoring interface for the goods-carrying vehicle's weight. The website is intended to display the weight measured by the load cell arrangement, allowing the user to observe the loading condition of the vehicle. The live weight monitoring feature makes it possible to view changes in the measured load without depending only on a physical display installed on the system.

The website also includes an overload-alert feature. The system uses a predefined weight threshold of 60 kg to identify an overload condition. When the measured weight exceeds this

limit, the overload condition is indicated through the website. This feature provides additional information to the user and helps make the loading condition easier to monitor.

The website forms the monitoring part of the overall system, while the STM32 microcontroller performs the main hardware control operations. Together, these parts support the objective of monitoring the load and providing an indication when the permitted weight limit is exceeded.

Python provides flexibility for developing and modifying the website according to the requirements of the project. The exact Python framework, database, hosting arrangement and communication protocol used in the implementation have not been specified and can be included once those details are confirmed.

Chapter 7

Simulation & Experimental Results

The project has been built on physical hardware. The following results are limited to the tests explicitly confirmed by the project team. No unreported numerical accuracy, response time, reliability, or environmental performance is claimed.

7.1 Confirmed Testing Status

Table 7.1: Confirmed hardware testing status

Function	**Status**
Live weight readings from the load-cell system on the website	Tested
Overload detection when weight exceeds 60 kg	Tested
Relay switching to prevent the vehicle from starting	Tested
Overload alert appearing on the website	Tested

7.2 Observed System Behaviour

The configured threshold is 60 kg. At a measured weight of 60 kg or less, the relay remains OFF and the vehicle can start. When the measured weight exceeds 60 kg, the relay switches to interrupt the line associated with the vehicle ECU and fuel pump, preventing the vehicle from starting. Live weight readings on the website, overload detection, relay switching, and the website overload alert have all been tested according to the project team's confirmation.

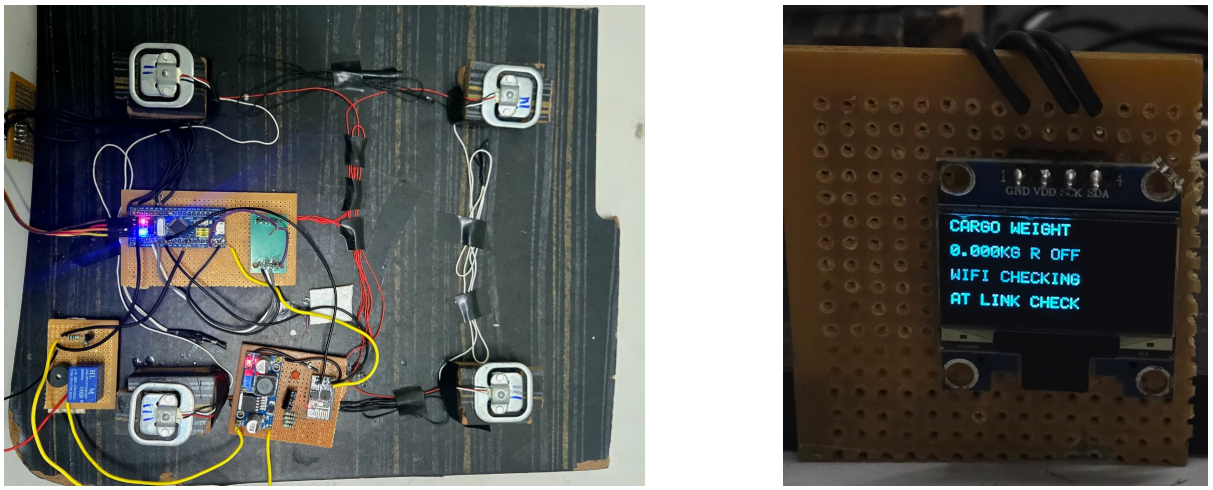


Figure 7.1: Hardware testing

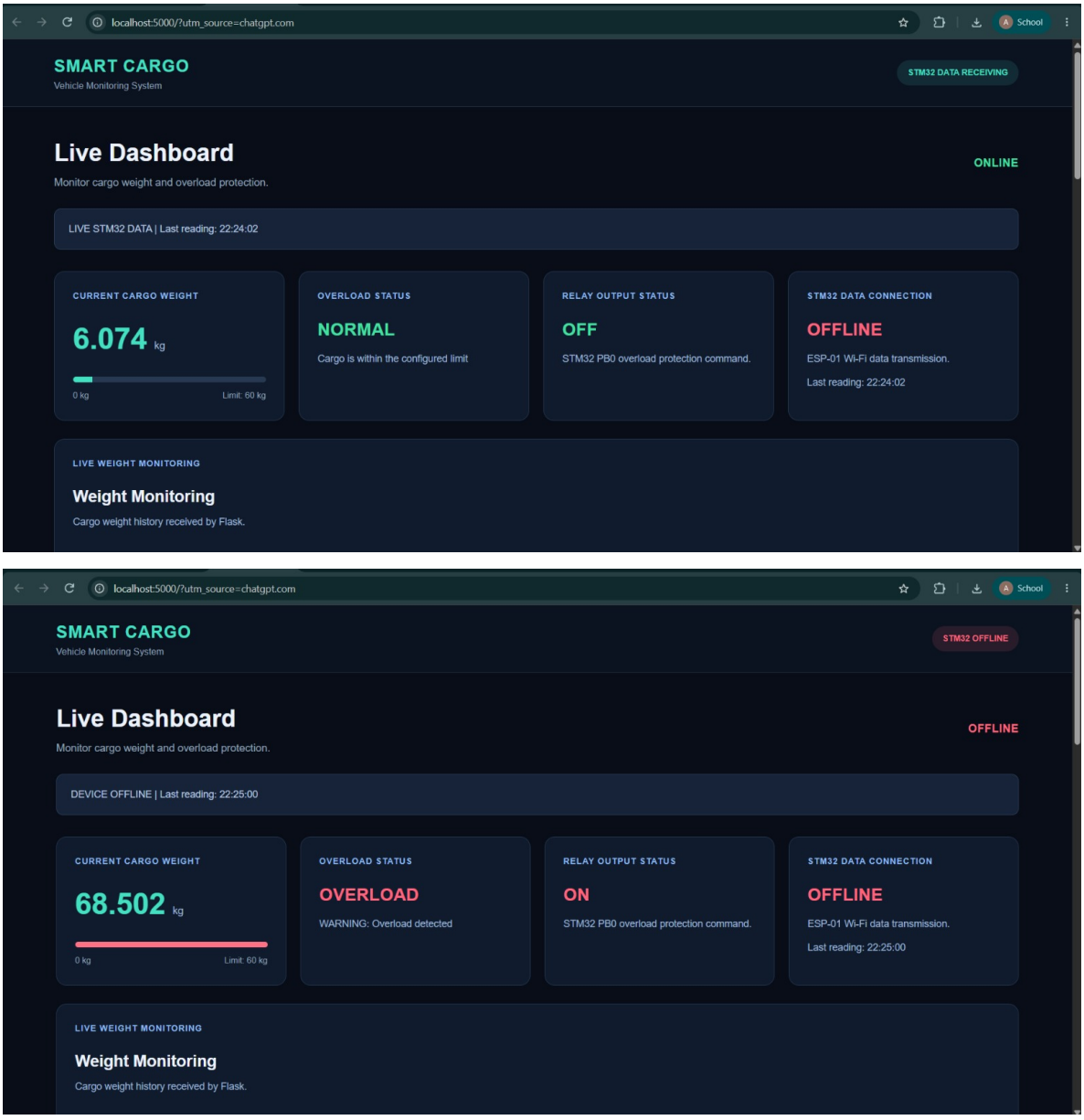


Figure 7.2: Software testing

Chapter 8

Conclusions

A microcontroller-based overload protection system for goods-carrying applications has been developed using four half-bridge load cells, an HX711 module, an STM32 Blue Pill microcontroller, a BC547 relay driver, a 5 V relay, and an ESP-01 Wi-Fi module. The load-cell arrangement measures the total load on the platform, and the controller compares the measurement with the configured threshold of 60 kg.

When the measured weight is 60 kg or less, the relay remains OFF and the vehicle can start. When the measured weight exceeds 60 kg, the relay switches the line associated with the vehicle ECU and fuel pump to prevent the vehicle from starting. The ESP-01 supports the transfer of live data to a Python-based website.

The project team confirmed that live weight readings on the website, overload detection above 60 kg, relay switching, and the website overload alert have been tested.

Future Work

Further improvements can be made to the proposed system to enhance its safety features and communication capabilities. A GSM module can be integrated to send warning messages to the vehicle owner or concerned authorities whenever an overload condition is detected. This will provide an alert even when the user is not monitoring the website.

A GPS module can also be interfaced with the STM32 microcontroller to obtain the vehicle's location. The location information can be combined with the overload alert and sent to the concerned authorities through the GSM module. This will help authorities identify the vehicle's location and take appropriate action when required.

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